

DESIGN & SIMULATION OF A

SMART WATER TANK

Overflow Protection & Pump Control System

Using Combinational Logic

6-module project presentation

Course: Digital Design & Computer Architecture

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System Inputs & Design Assumptions



Symbol	Input meaning	Logic 1 means
L	Low-level sensor	Water is at/above the low sensor point
M	Medium-level sensor	Water is at/above the medium point
H	High-level sensor	High-level limit is reached
P	Pump output	1 = ON, 0 = OFF

Assumption: sensors are ordered vertically (L below M below H). Real hardware needs sensor conditioning and electrical isolation.

Water-Level Sensing Concept



- Three digital sensors convert a changing water level into binary signals.
- L — LOW: indicates the tank has reached the minimum sensor point.
- M — MEDIUM: provides an intermediate water-level indication.
- H — HIGH: signals the upper limit; the pump must be OFF.
- Sensor polarity and exact thresholds must be confirmed for the selected hardware.

What Is Combinational Logic?



L, M, H
Inputs

BOOLEAN LOGIC

P
Output

- The output depends only on the current inputs; no clock or stored state is required.
- When an input changes, the output updates after a small circuit delay.
- A practical pump system may add a latch, timer or interlock.

Truth Table — Pump Decision



Chosen rule: pump ON only when the tank is below the low-level point.

L	M	H	P	Interpretation
0	0	0	1	Below low sensor — fill
0	0	1	0	Invalid sensor combination
0	1	0	0	Invalid sensor combination
0	1	1	0	Invalid sensor combination
1	0	0	0	Low sensor reached
1	0	1	0	Invalid sensor combination
1	1	0	0	Medium level reached
1	1	1	0	High limit reached

This demonstration turns the pump OFF as soon as L becomes 1, so it does not continue filling to the medium level.

Simulation Setup



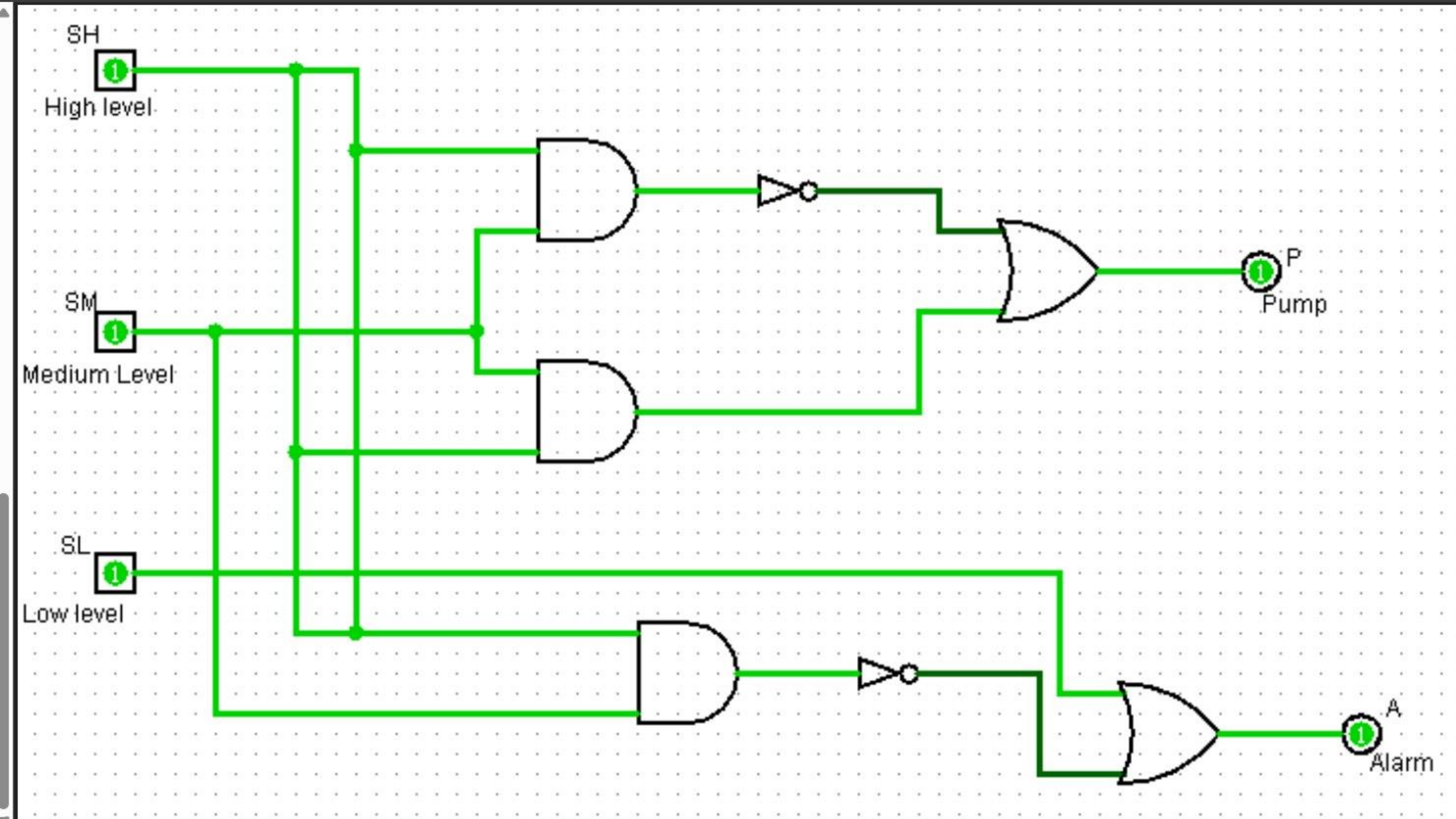
- Suggested tools: Logisim Evolution or Digital logic simulator.
- 1. Place three input pins labelled L, M and H.
- 2. Add a 3-input NOR gate or an equivalent gate network.
- 3. Connect P to an output pin or LED indicator.
- Toggle all eight input combinations, compare each output with the truth table, then save the circuit and capture a screenshot.

CONCLUSION



- Digital sensors represent water level using binary inputs.
- The truth table defines the pump decision clearly.
- P = ¬(L + M + H), implemented by a NOR gate.
- Simulation checks every input combination before hardware implementation.

Design • Simulate • Verify • Improve


SH
1
High level
SM
1
Medium Level
SL
1
Low level
P
1
Pump
A
1
Alarm



Thank You!

For your attention